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Preliminary Nozzle Design for use in a Small-Scale, High Mach Number Wind Tunnel

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Nomenclature

A	$Local\ Area$	m^2
A_t	$Throat\ \ Area$	m^2
a	$Speed\ of\ Sound$	$\frac{m}{s}$
C^+	$Right\ Running\ Characteristic$	
C^-	$Left\ Running\ Characteristic$	
c_p	$Constant\ Pressure\ Heat\ Capacity$	$\frac{J}{kgK}$
ϵ	$Area\ Ratio$	
F	$Thrust$	N
γ	$Ratio\ of\ Specific\ Heats$	
M	$Mach\ Number$	
μ	$Mach\ angle$	$degrees$
$\dot{m}$	$Mass\ Flow\ Rate$	$\frac{kg}{s}$
ν	$Prandtl - Meyer\ Angle$	$degrees$
P	$Local\ Pressure$	Pa
P_t	$Stagnation\ Pressure$	Pa
ϕ	$Local\ Angle\ of\ Characteristic\ Line$	$degrees$
$\dot{q}$	$Heat\ Transfer\ Rate$	W
R	$Gas\ Constant$	$\frac{J}{kgK}$
T	$Local\ Temperature$	K
T_t	$Stagnation\ Temperature$	K
θ	$Flow\ Turning\ Angle$	$degrees$
u	$x\ Direction\ Velocity\ Component$	$\frac{m}{s}$

V	$Velocity\ Magnitude$	$\frac{m}{s}$
v	$y\ Direction\ Velocity\ Component$	$\frac{m}{s}$
x	$x\ Spatial\ Location$	m
y	$y\ Spatial\ Location$	m

Executive Summary

A preliminary design and analysis of a converging-diverging nozzle was performed. The nozzle is to be used in a small-scale, high-Mach number wind tunnel for the purpose of material characterization. The diverging contour was generated using the method of characteristics and an evaluation was carried out using numerical simulation. The feasibility of the nozzle design and supporting system elements was determined through a review of existing high-Mach number wind tunnel facilities. A brief investigation into the supporting elements of the system was performed to facilitate the continuation of the project.

Introduction

Motivation

High-Mach number wind tunnels (HWTs) are a necessary component of experimental facilities that the Department of Energy and Department of Defense require to carry out their national security programs and in their collaborations with other agencies including NASA or the aircraft and space industries [1]. HWTs allow for experimental testing and characterization of materials in high-energy flows and under extreme conditions, such as those experienced by a NASA space vehicle's atmospheric reentry [2-6]. Our collaborators at Lawrence Livermore National Laboratory (LLNL) believe surface material response characterization can just as well be accomplished with the investigation of local effects on a small coupon sample, rather than a full vehicle model. The coupon test requires only a laboratory tabletop, compact wind tunnel. A small-scale HWT approach is more economical, provides for rapid testing and lends itself more readily to high fidelity diagnostics.

Background

Traditionally, HWT facilities have been large, building-sized facilities, that are costly to run, with large-scale models, and do not allow for rapid testing of materials [7]. The facility at NASA Langley is depicted in Figure 1.

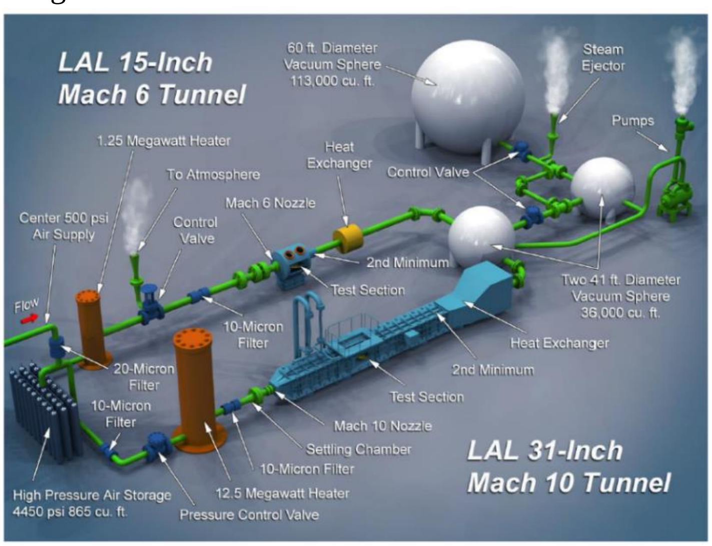

Figure 1. Mach 6 and Mach 10 Tunnels at NASA Langley. From [7]

Recently, as interest in high Mach number flight has increased, several universities have constructed their own HWTs [8-10]. The two facilities known as ACT-I and ACT-II at the University of Notre Dame and University of Illinois at Urbana-Champaign, respectively, are of interest due their relatively small size and high-Mach number performance. Images of the two facilities can be seen in Figures 2 & 3. While still large, these tunnels are considered "laboratory" sized.

Figure 2. HWT ACT-I Facility at University of Notre Dame. From [9]

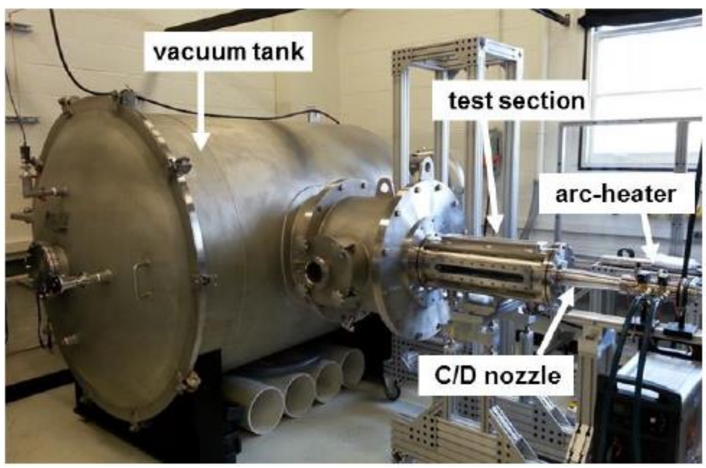

Figure 3. HWT ACT-II Facility at University of Illinois at Urbana-Champaign. From [10]

It is important to note that these two tunnels were designed for combustion research and the large tanks downstream of the test section are to contain the combustion products. The nozzle design is the focus of the work herein and are of interest the nozzle parameters of ACT-I summarized in Table 1 (adapted from [9]).

Table 1. Parameters of ACT-I contoured nozzles [9]

Mach number	Throat diameter	Exit diameter	Core diameter	Divergent length	Max. test time
4.5	12.6 mm	60.5 mm	~40 mm	255.5 mm	1 s
6	7.8 mm	67.6 mm	~44 mm	383.2 mm	1 s
9	6.4 mm	181 mm	~90 mm	681.3 mm	0.4 s

The main design parameter of interest is the divergent length which we believe is short enough to permit the remaining components to fit on a laboratory tabletop (~2-3m). Another point of interest is that for the Mach 9 nozzle the test time is limited by the pressure rise in the vacuum chamber, whereas the Mach 4.5 and 6 nozzles have run times limited by the temperature of the arc-heater electrodes [9].

Scope

This project is intended to be a preliminary design study and, as such the scope includes the following tasks:

- Develop a process for creating the appropriate nozzle geometry using the method of characteristics.
- Analyze the flow field for the generated nozzle geometry using simulation analysis.
- Outline additional elements of the wind tunnel system

A desired Mach number range in the test section of $3 \leq M_{test} \leq 12$ is the driving parameter of the system design specifications. In addition, we wish to keep the total length from nozzle inlet to exit to a minimum with the goal being that the full system design fits on a laboratory tabletop (~2-3m in length). The requirements for the wind tunnel are a challenge due to the extreme flow conditions required to create the high Mach number at the test location. For high Mach number flows ($M \geq 6$) the temperature, pressure, and area change across the nozzle are very large and material limitations become an important factor. Additionally, the thermodynamic changes that the flow experiences require that the supply gas be free of moisture or other impurities that could condense during the expansion.

Nozzle

Fundamentals

The acceleration of a flow from subsonic to supersonic velocities differs from purely subsonic flow. An isentropic analysis of mass and momentum conversation results in what is known as the Mach-area relation.

$$\frac{dA}{A} = -\frac{dV}{V}(1 - M^2) \tag{1}$$

For subsonic flow, $M < 1$, a decrease in area accompanies an increase in velocity whereas supersonic flow requires an increasing area to increase the fluid velocity. This relationship dictates that to accelerate a fluid to supersonic velocities it must first pass through a converging section until sonic velocity is obtained, then pass through a diverging section until the desired velocity is obtained. This configuration is known as a converging-diverging nozzle.

The nozzle analysis began with an isentropic analysis using the equations for total temperature, T_t, and pressure, P_t,

$$\frac{T_t}{T} = 1 + \frac{\gamma - 1}{2} M^2 \tag{2}$$

$$\frac{P_t}{P} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\frac{\gamma}{\gamma - 1}} \tag{3}$$

derived from the steady, one-dimensional, adiabatic energy balance with Mach number, M, and ratio of specific heats, γ, as shown in Figure 4 for air and argon.

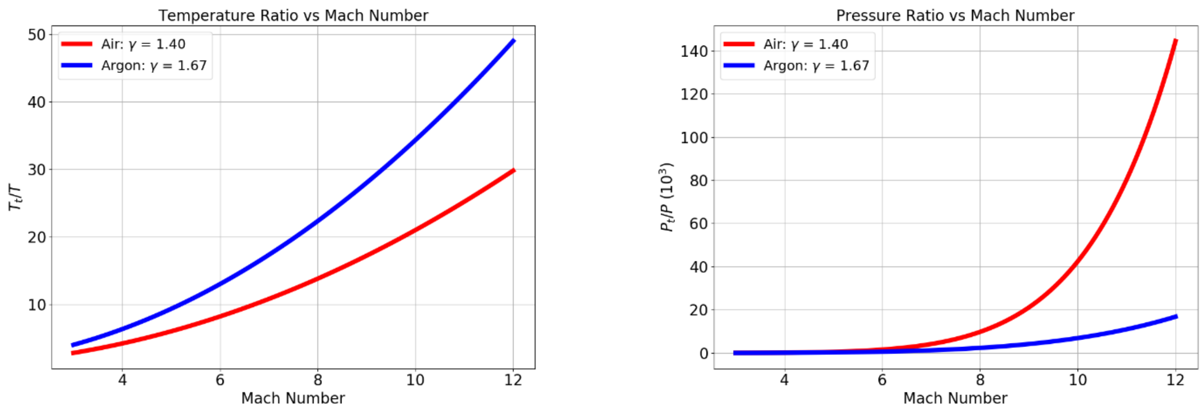

Figure 4. Exponential Increase of Temperature and Pressure ratios with Mach number

The large temperature ratios necessary at higher Mach numbers for both air and argon indicate the need for a large heating system to keep the gasses from cooling to their liquification temperature during expansion ($T_{l_{Ar}} = 87K$ & $T_{l_{Air}} = 79K$). The required pressure ratio favors the use of argon when $M \geq 7$. However, an argon system would require more heating as shown by the temperature ratio in Figure 4. Next, the area ratio, per Equation 4 (below), was calculated over a range of Mach numbers for air and argon. Results are shown in Figure 5.

$$\epsilon = \frac{A_2}{A_t} = \sqrt{\frac{2}{\gamma + 1}\left(1 + \frac{\gamma - 1}{2} M_2^2\right)^{\frac{\gamma+1}{\gamma-1}}} \tag{4}$$

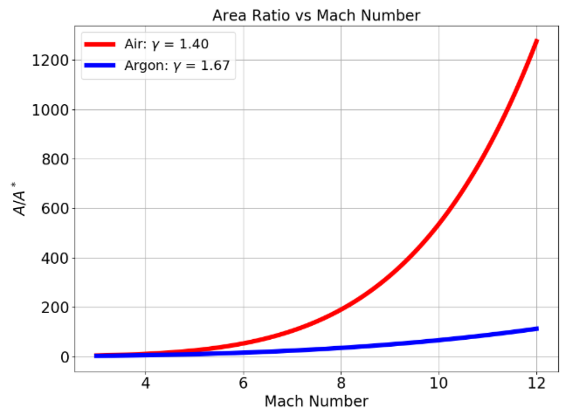

Figure 5. Sensitivity of area ratio to γ at high Mach number

At the higher Mach numbers, the area ratio becomes quite exaggerated for air, whereas for argon the area ratio is smaller and would allow for a larger throat region for a fixed exit area. Nozzles designed with a fixed exit area are common practice for wind tunnels as it allows for multiple nozzles to fit the same test section. However, with a fixed exit area, as the Mach number increases the throat size will decrease which could lead to problems. For example, the elevated temperatures combined with the reduced throat area will increase the local heat flux to the nozzle walls. The increased heat flux will require additional cooling to maintain an operating temperature that allows the throat walls to main their structural integrity. Additionally, as the throat size decreases there will be an increased sensitivity to manufacturing tolerances. As the relative size of the surface roughness becomes larger, there is the risk of the flow quality being degraded which will propagate downstream. The nonuniform surface features will also serve as hotspots along the wall where the local temperature might exceed a safe operating temperature as the flow is decelerated and heated to stagnation temperature.

The nozzle will be producing thrust which must be accounted for when designing the structural elements of the system to prevent system damage. The generated thrust from a nozzle, assuming an optimal expansion process, is given by Equation 5.

$$F = \dot{m}V_2 \tag{5}$$

Here the mass flow rate, $\dot{m}$, and velocity at the nozzle exit, v_2, are found by Equations 6 & 7, respectively.

$$\dot{m} = \frac{P_1 A_t \gamma \sqrt{\left(\frac{2}{\gamma - 1}\right)^{\frac{\gamma+1}{\gamma-1}}}}{\sqrt{\gamma R T_1}} \tag{6}$$

$$V_2 = \sqrt{\frac{2\gamma}{\gamma - 1} R T_1 \left(1 - \left(\frac{P_2}{P_1}\right)\right)^{\frac{\gamma-1}{\gamma}}} \tag{7}$$

An exit area of 50cm^2 was arbitrarily chosen as an example to allow for calculation of the throat area and the thrust. The pressure and temperature of the test section conditions corresponded to an altitude of 30km. Thrust was then plotted over a range of Mach numbers, shown in Figure 6.

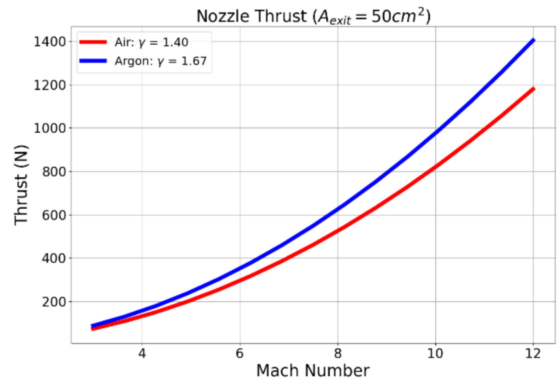

Figure 6. Expected thrust

There are only minor differences in the expected thrust between the two gases and the maximum expected values are low enough that a standard bolt support for the nozzle can withstand the shear force [11]. The variation of thrust during off design performance is minimal by inspection of (6) & (7).

Planar vs Axisymmetric

There are two standard nozzle shapes for use in wind tunnels, planar and axisymmetric. Planar or 2-D nozzles have two contoured walls and two parallel walls with expansion occurring in only one dimension. This configuration results in a rectangular cross section throughout which is more suited to a wind tunnel test section where a uniform flow profile is desired. Planar nozzles are easier to manufacture and can be modified to generate different Mach numbers. However, at higher Mach numbers ($M > 6$), the throat experiences significant thermal and mechanical stresses [12]. Axisymmetric nozzles have a circular cross section throughout and experience expansion in two dimensions. They are more complicated to manufacture but the symmetrical shape facilitates cooling of the throat. If a rectangular test section is used the transition from a circular to rectangular cross section must be accounted for to understand the impact on flow quality.

Method of Characteristics

Theory

The nozzle must produce uniform flow to allow for accurate material characterization with known and undeviating conditions. To achieve uniformity at the exit, the nozzle must be shock-free and have walls parallel to the flow direction at the exit plane. There are two approaches to nozzle design, direct design where the wall contour is the final output from a single computation or design-by-analysis which involves optimization using flow simulation tools [12]. A direct design approach was chosen for this project due to its lower investment and ease of use. The method of characteristics is the most common direct design method and was used here. The method of characteristics, first developed by Prandtl and Busemann [13], utilizes expansion via

Mach waves to accelerate the flow from sonic to supersonic while also turning the wall contour in such a way that any generated waves are cancelled out before they can coalesce into shocks. A brief overview is provided below, and the complete derivation can be found in most standard compressible flow or gas dynamics textbooks [14, 15].

When supersonic flow expands through a convex corner, a series of Mach waves are generated that gradually turn the flow, the grouping of these waves is known as a Prandtl-Meyer expansion fan, shown in Figure 7.

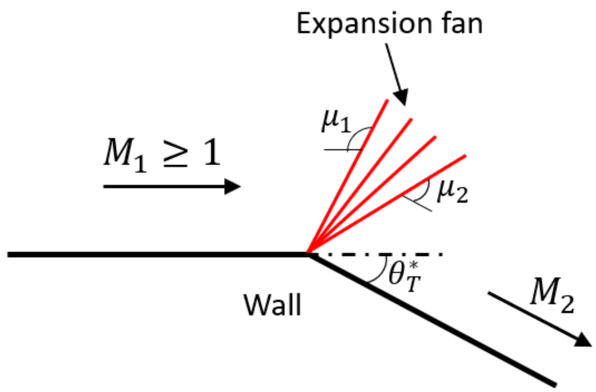

Figure 7. General expansion fan at a convex corner

The Mach angle, μ, is related to the Mach number by Equation 8.

$$\mu = sin^{-1}\frac{1}{M} \tag{8}$$

If we then consider a single Mach wave with expansion angle, $d\nu$, the incremental increase in velocity can be found by

$$\frac{dV}{V} = tan\,\mu\,d\nu \tag{9}$$

The angle that the flow must be expanded through to reach a specified Mach number, starting from $M_1 = 1$, is found by integrating the velocity relationship

$$\frac{dV}{V} = \frac{1}{\sqrt{M^2-1}}d\nu \tag{10}$$

which results in

$$\nu = \sqrt{\frac{\gamma+1}{\gamma-1}}tan^{-1}\left(\sqrt{\frac{\gamma-1}{\gamma+1}(M^2-1)}\right) - tan^{-1}\sqrt{M^2-1} \tag{11}$$

Equation 11 is known as the Prandtl-Meyer function (PMF). The relation between M_1, M_2 and θ_T^* then becomes

$$\nu(M_1) = \nu(M_2) + \theta_T^* \tag{12}$$

Now that we have established the relation between the Mach number and wall angle, a brief description of characteristics is warranted. A characteristic line is defined as a line on which there is a discontinuity in the velocity gradient [15]. Starting from the potential equation for isentropic, irrotational, steady flow of a perfect gas with constant specific heats, for a non-unique solution velocity derivative to exist, the following equation must hold [14].

$$C_1 \left(\frac{dy}{dx}\right)^2 - 2C_2 \frac{dy}{dx} + C_3 = 0 \tag{13}$$

where C_1, C_2, and C_3 are found to be

$$C_1 = a^2 - u^2 \tag{14}$$

$$C_2 = -uv \tag{15}$$

$$C_3 = a^2 - v^2 \tag{16}$$

Solving (13) results in

$$\frac{dy}{dx} = \frac{-uv \pm \sqrt{a^2u^2 + a^2v^2 - a^4}}{a^2 - u^2} \tag{17}$$

Therefore, at each point in the flow there exists two characteristic lines with slopes given by

$$\left(\frac{dy}{dx}\right)^- = \frac{\left(-\frac{uv}{a^2}\right) + \sqrt{M^2 - 1}}{1 - \left(\frac{u^2}{a^2}\right)} \tag{18}$$

$$\left(\frac{dy}{dx}\right)^+ = \frac{\left(-\frac{uv}{a^2}\right) - \sqrt{M^2 - 1}}{1 - \left(\frac{u^2}{a^2}\right)} \tag{19}$$

However, by representing the velocity in radial coordinates and using (8), the slopes can be simply written as

$$\left(\frac{dy}{dx}\right)^- = tan(\theta - \mu) \tag{20}$$

$$\left(\frac{dy}{dx}\right)^+ = tan(\theta + \mu) \tag{21}$$

which shows that the characteristic lines are inclined by the Mach angle to the flow direction, Figure 8.

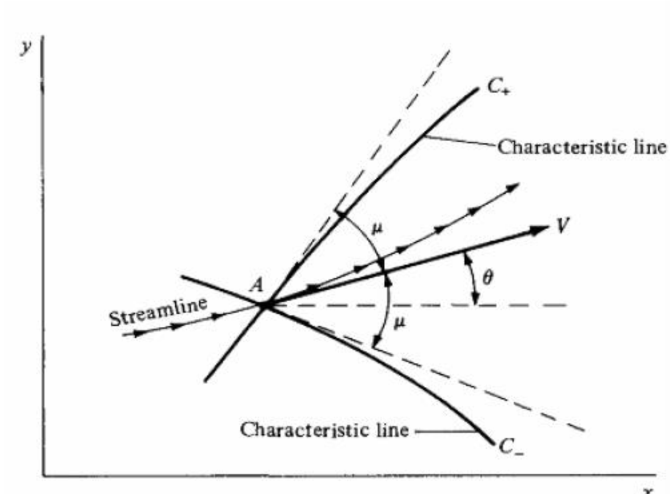

Figure 8. Characteristic Lines through point A. From [14]

An expression for the velocity is still required to solve the flow problem. Utilizing the expression [14]

$$\frac{dv}{du} = \frac{1 - uv \pm a\sqrt{u^2 + v^2 - a^2}}{v^2 - a^2} \tag{22}$$

We ultimately reach the conclusion that

$$\left(\frac{dv}{du}\right)^- = -\frac{1}{\left(\frac{dy}{dx}\right)^+} \tag{23}$$

$$\left(\frac{dv}{du}\right)^+ = -\frac{1}{\left(\frac{dy}{dx}\right)^-} \tag{24}$$

It can be further shown that

$$\frac{1}{V}\frac{dV}{d\theta} = \mp tan\,\mu = \mp\frac{1}{\sqrt{M^2 - 1}} \tag{25}$$

We can then see that (25) is in the same form as (10), however, a distinction must be made that (9) is integrated *across* the wave whereas (25) is integrated *along* the characteristic [14]. Using the previous results of (11) we arrive at

$$\theta + \nu = constant = C^+ \tag{26}$$

$$\theta - \nu = constant = C^- \tag{27}$$

Now that we have a simplified relationship between the Mach number and flow angle along a characteristic line, the calculation procedure on a point-by-point basis is described. For an internal point, shown in Figure 9, we assume full knowledge of the states of points 1 and 2 and use (26) and (27) to arrive at

$$\theta_3 = \frac{1}{2}[(C^-)_1 + (C^+)_2] \tag{28}$$

$$\nu_3 = \frac{1}{2}[(C^-)_1 - (C^+)_2] \tag{29}$$

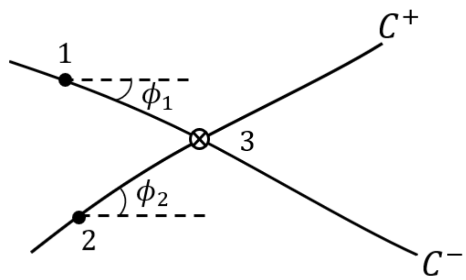

Figure 9. Characteristic lines for an internal point

The location of point 3 is determined by approximating the characteristics as locally straight lines. The equations for the coordinates of point 3 are then

$$x_3 = \frac{x_2\, tan(\phi_2) - x_1\, tan(\phi_1) + y_1 - y_2}{tan(\phi_2) - tan(\phi_1)} \tag{30}$$

$$y_3 = (x_3 - x_1)\, tan(\phi_1) + y_1 \tag{31}$$

$$\phi_1 = \frac{1}{2}(\theta_1 + \theta_3) - \frac{1}{2}(\mu_1 + \mu_3) \tag{32}$$

$$\phi_2 = \frac{1}{2}(\theta_2 + \theta_3) + \frac{1}{2}(\mu_2 + \mu_3) \tag{33}$$

For a wall point, as shown in Figure 10,

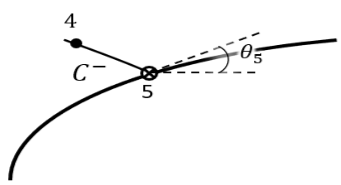

Figure 10. Characteristic line of a wall point

the turning angle is now known which leaves ν_5 as the only unknown which is found by

$$\nu_5 = \nu_4 + \theta_4 - \theta_5 \tag{34}$$

The point location is found using equations (30) and (31) with angles now given by

$$\phi_1 = \frac{1}{2}(\theta_5 + \theta_{wall}) \tag{35}$$

$$\phi_2 = \frac{1}{2}(\theta_5 + \theta_4) + \frac{1}{2}(\mu_5 + \mu_4) \tag{36}$$

Where θ_{wall} is the turning angle at the previous wall location.

Minimum Length vs. Gradual Expansion

Having established the basis of the method of characteristics, the application to supersonic nozzle design can now be explored. There are two types of supersonic nozzles, gradual expansion and minimum length. The gradual expansion nozzle, shown in Figure 11, expands the flow through a series of waves from point *a* to *c* as the wall angle increases up to the maximum given by

$$\theta_{w_{max}} = \frac{\nu_M}{2} \tag{37}$$

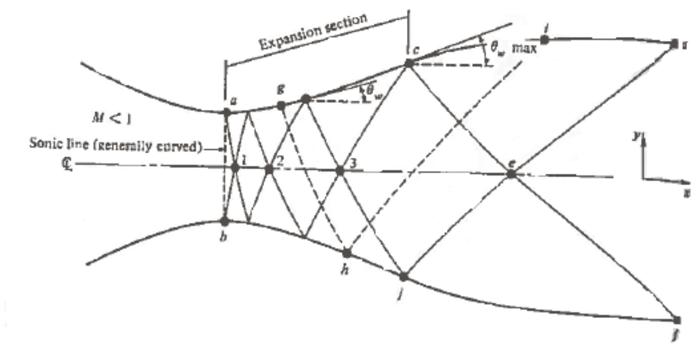

Figure 11. Gradual expansion nozzle. From [14]

where ν_M is the value of the PMF associated with the design exit Mach number. Once $\theta_{w_{max}}$ is reached the wall angle begins to decrease back to 0° to cancel the waves. General expansion nozzles are longer than minimum length nozzles but produce higher quality flow especially at higher Mach numbers where the wall angle can become quite large ($\theta \geq 40°$).

For minimum length nozzles, shown in Figure 12, the expansion section, *ac*, is shrunk to a single point and the expansion occurs across a Prandtl-Meyer wave centered at the sharp corner of the throat (point *a*).

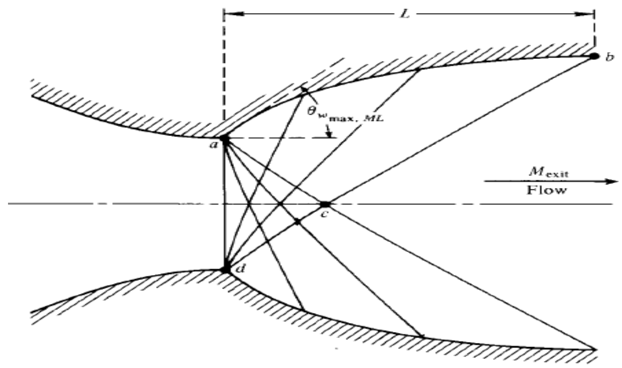

Figure 12. Minimum length nozzle. From [14]

Minimum length nozzles are commonly found in rocket motors where the length and weight reduction are critical [15].

Many method of characteristic codes can be found on open-source code repositories, however, each code has variations depending on their specific application. It was found that available codes were inadequate for the purposes of this project. Therefore, a method of characteristics code was written to generate minimum length nozzle contours. The code was verified through comparison with an example problem in the literature [14]. The implementation of the method of characteristics is simpler for minimum length nozzles and within the scope of this work is adequate for preliminary design and analysis but as the design is refined, modifications will need to be made to produce gradual expansion contours [16, 17]. To use the code, the user must specific an exit Mach number, a ratio of specific heats, the number of characteristics, and an initial angle, θ_i. Two of the assumptions used in the application of the method of characteristics were to replace the infinite number of waves with a finite number of characteristics lines and assume the initial angle is small. The effect of these assumptions in the nozzle quality was examined for a Mach 3 nozzle using $\gamma = 1.4$. The area under the contour was used to quantify the quality of the nozzle and was numerically calculated using the trapezoid rule. Two cases were considered. The first, increased the number of characteristics while fixing the initial angle at a small value of 1e-10°, shown in Figure 13.

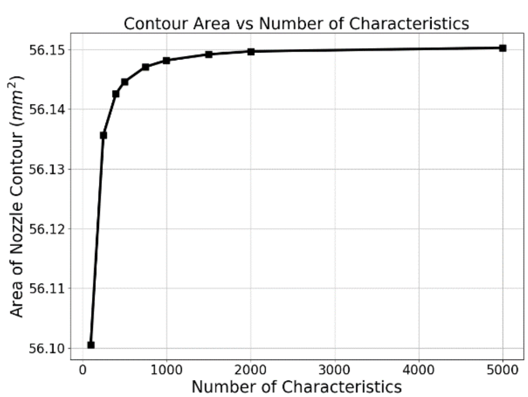

Figure 13. Contour area asymptotically increasing with increased number of characteristics

The area change of the contour begins to level off at around $n = 1000$ and it is worth noting that the change in area between $n = 100$ & $n = 5000$ is less than 0.1% indicating that further increases in the number of characteristics has a negligible effect on the nozzle shape. The second, decreased the initial angle while using a fixed number of characteristic lines ($n = 400$), shown in Figure 14.

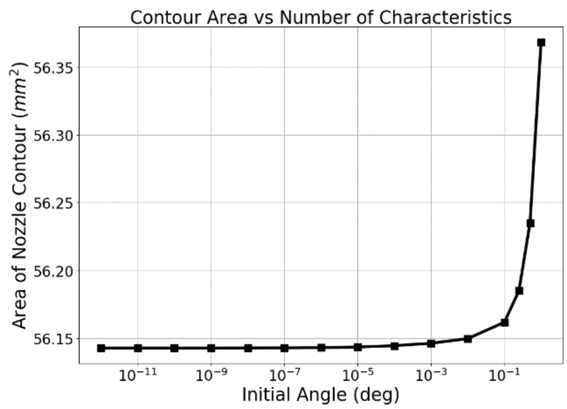

Figure 14. Asymptotically decreasing contour area with decreasing initial angle, 400 characteristic lines

This trend is satisfactory to visually conclude that using an initial angle below 1e-5° has a negligible effect on the nozzle profile. It is again worth noting that the initial angle spans 13 orders of magnitude and the percentage change in nozzle area is only 0.4%. A sample output from the method of characteristics code is shown in Figure 15.

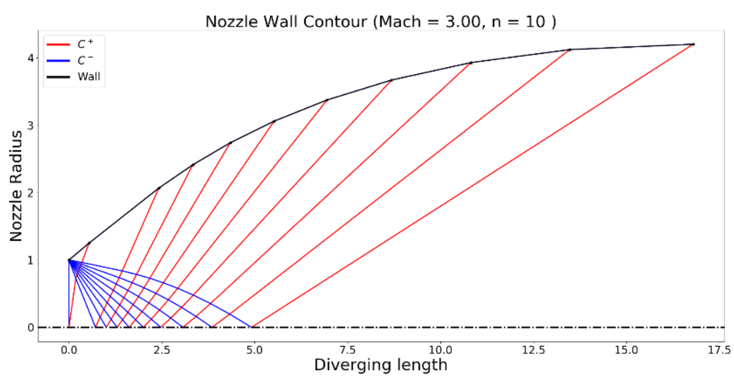

Figure 15. Diverging contour for a Mach 3 nozzle using 10 characteristics

This contour was produced using 10 characteristic lines (for visual clarity) and an initial θ of $10^{-10°}$.With this few characteristics the discontinuities in the nozzle wall are still pronounced but for larger numbers of characteristics, $n > 250$, these discontinuities do not affect the flow quality as later seen in the simulation analysis. The nozzle contours considered in the following section used $\theta = $ 1e-10° & $n = 400$ characteristics, above this number of characteristics the CAD program had a sharp decrease in performance.

Numerical Flow Simulation

After establishing a method of nozzle design, analysis using flow simulation analysis was performed to verify that nozzles were producing the expected Mach number at the exit. The contour generated from the method of characteristics was exported as a set of coordinate pairs which could then be imported into a CAD suite. For this work, Autodesk Inventor® was the CAD tool used [18]. The contour design only accounted for the diverging section of the nozzle and to accurately judge performance only this domain was analyzed. Attempts to include a converging section resulted in shocks as the design method does not account for a curved sonic line [14]. The coordinates were imported and connected by a spline, vertical faces were added at the throat and exit planes, and the profile was then extended into the z-axis. This geometry is symmetrical about the z-axis and y-axis and corresponds to one quadrant of the planar nozzle and is shown in Figure 16.

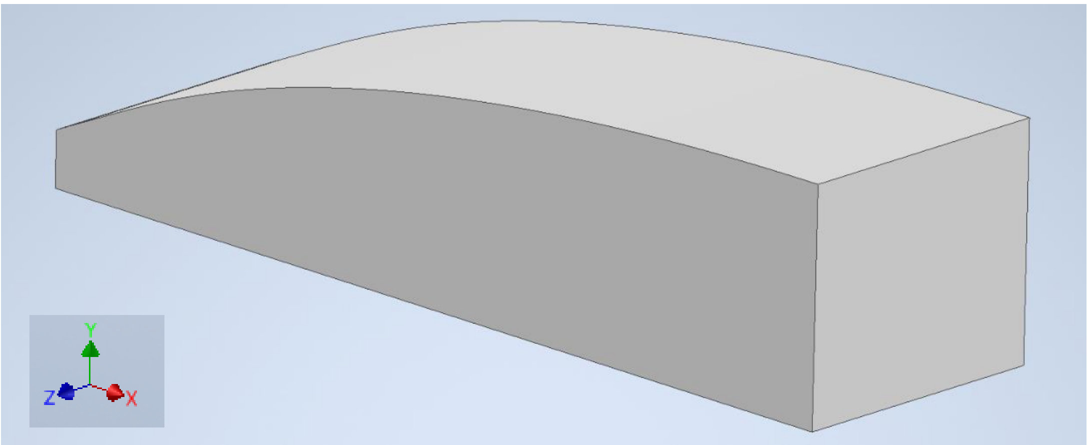

Figure 16. Nozzle geometry, upper right quadrant

The geometry was then imported into ANSYS® Fluent® for analysis [19]. The built-in meshing capabilities were used to construct the mesh. A structured mesh was chosen to reduce the number of elements needed and to reduce computational time. A representative mesh is shown in Figure 17.

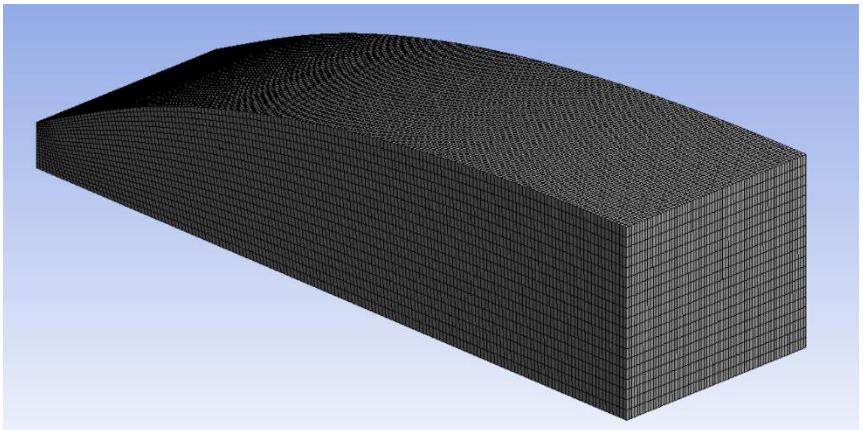

Figure 17. Structured mesh with 238k elements

An inviscid solution method was chosen to correspond with the assumptions made in the method of characteristics formulation. The addition of viscous effects requires iteration of the nozzle contours and was beyond the scope of this work. The simulation required four boundary conditions for the inlet, outlet, walls, and the symmetry plane. The inlet conditions were set at $P_{static} = 19.409$ atm, $T_{static} = 700$ K, and $M = 1$. The outlet conditions were set at $P_{static} = 1$ atm and $T_{static} = 300$ K. These pressure and temperature values are related by equations (2) & (3). The walls were adiabatic and stationary. Symmetry was applied to the $+z$ & $-y$ faces.

To determine if the solution is independent of the grid three meshes were considered with 104k, 238k, and 1.16M elements. Two locations of interest were analyzed, the exit plane and the centerline, to determine solution convergence. First, the Mach number in the exit plane was examined, shown in Figure 18.

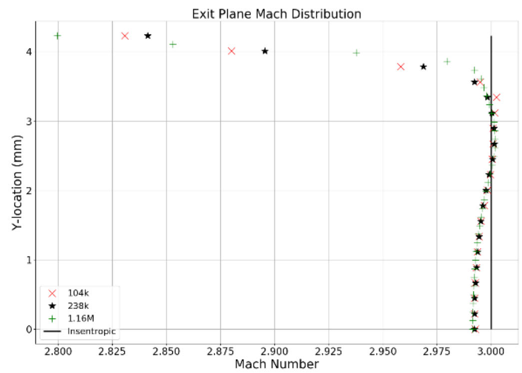

Figure 18. Exit plane Mach distribution for mesh sizes of 104k, 238k, and 1.16M elements

The Mach number distributions are unchanging over the refinement of the mesh indicating the solution is converged. Additionally, the distributions show close agreement with the expected isentropic value. Next, the pressure distribution along the centerline was examined, shown in Figure 19.

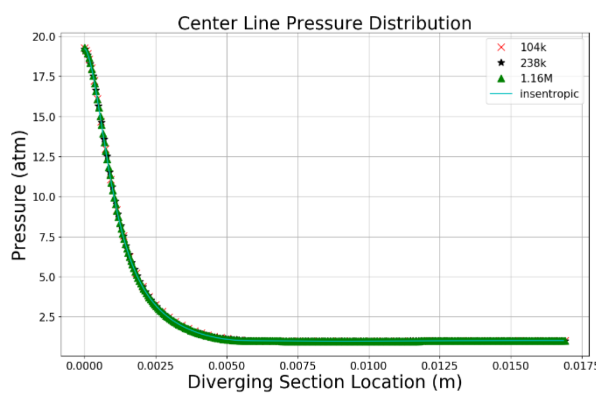

Figure 19. Centerline pressure distribution for mesh sizes of 104k, 238k, and 1.16M elements

Again, the solutions from the three meshes coincide with each other throughout the domain and follow the expected isentropic distribution. These two results give confidence that the correct solution is being obtained. The remaining results shown are from the 238k element case.

The overall Mach distribution was also examined to ensure there were no irregularities. The Mach contour along the z-direction symmetry plane is shown in Figure 20.

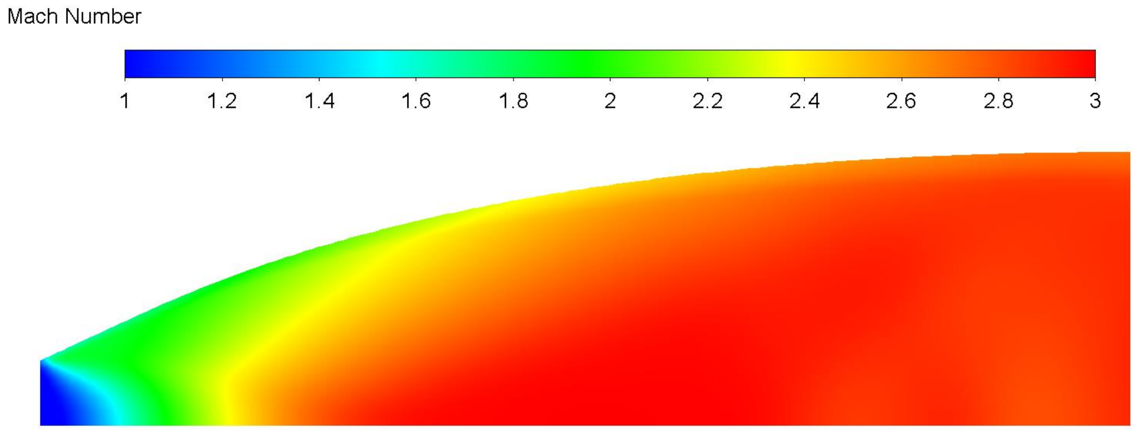

Figure 20. Inviscid Mach contour of diverging section

This distribution is as expected. The centerline location experiences the most rapid acceleration, ultimately reaching the design Mach number the soonest, while the near-wall region requires the entire length.

System Elements

While this work was focused on the nozzle design and analysis, other elements of the wind tunnel system warrant a brief overview which can serve as a starting point for the continuation of the project.

Gas Supply

The working fluid for the wind tunnel can be provided by two sources, atmospheric air or compressed gas cylinders. Atmospheric air would require filtration and desiccation before being compressed to the operating pressure. While this would allow for potentially longer operation, the variability of air composition could lead to problems with experimental reproducibility. Additionally, a high-powered compressor would add an additional costs and complexity to the wind tunnel. Compressed gas cylinders would provide greater consistency and control over the quality supplied. Furthermore, if the wind tunnel is to operate with an inert gas, such as argon, a compressed cylinder would be the only option.

Heater

The gas must be heated before expansion to prevent condensation during expansion. A quick estimation of the heat input required for Mach 10 flow is found by using a steady-state energy balance across the heater section shown in Figure 21 leads to Equation (38).

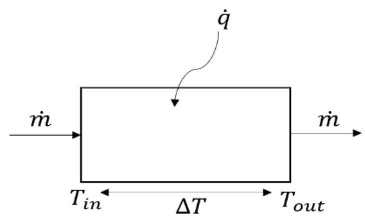

Figure 21. Steady-state energy balance

$$\dot{q} = \dot{m} c_p \Delta T \quad (38)$$

For a Mach 6 &10 nozzle moving air at ~1kg/s and with an exit temperature of 230K (corresponding to an altitude of 30km), the energy that must be supplied is on the order of 1.5MW and 5MW, respectively. There are multiple possible methods of heating the flow. These include, resistive heating, combustion heating, and arc heating. Resistive heating was found to be inadequate to provide the necessary energy for the expected mass flow rates within our size requirements [9]. Combustion heating could provide the required energy for the lower range of Mach numbers, but as the Mach number increases, the heat addition will become insufficient [20,21]. Additionally, with combustion heating, the flow would be contaminated by the combustion products that alter the ratio of specific heats, thus changing the necessary contour and the presence of water would lead to condensation problems [20]. To reach the necessary enthalpies, an arc heater is recommended [9,10]. An arc heater works by applying a high-voltage difference between two electrodes that experiences an electric breakdown which produces a high-current arc. The arc temperature rises from high levels of Ohmic heating occurring in the plasma column which mixes with the rest of the flow, drastically increasing the fluid temperature.

Test Chamber

The test chamber is where the material test samples are located, and we want optical access at this location for diagnostics. The chamber should be a sufficient size to allow the samples to be repositioned as needed. The chamber will hold a partial vacuum and the walls need to be thick enough to withstand the force and prevent any fatigue damage over the material life.

Diagnostics

Diagnostics for the wind tunnel will be provided from multiple non-intrusive sources. To perform material characterization the freestream must first be accurately measured. One method to accomplish this uses Particle Image Velocimetry (PIV) [22]. PIV works by seeding the flow field with fluorescent particles which can be imaged by a laser system to determine the flow speed and direction [22]. Another method is to use Rayleigh scattering measurements which uses the difference between incident laser frequency and the peak in the Rayleigh spectrum to calculate velocities [23]. To understand the shape and formation of the boundary layer over the material Schlieren imaging can be used. Schlieren imaging visualizes the density gradients in the flow that reveal the size and structure of large gradients such as those found in the boundary layer or across a shock [24]. The important parameters at the surface of the material samples are temperature, pressure, shear, and surface deformation. Temperature and pressure can be measured by means of temperature sensitive paint or pressure sensitive paint that use variations in a material’s luminescence to map gradients [25]. Shear measurements can be made using global interferometry skin friction which correlates the shear of a thin film of oil on the surface to that shear experienced by the material [26].

Downstream Elements

The components of the system downstream of the sample, consist of a diffuser, a reservoir, and a pump. The supersonic diffuser is a device used to slow the flow back to subsonic speeds in such a way that the total pressure loss is minimized, which increases the overall efficiency of the system. After the flow has slowed through the diffuser it will enter a reservoir. During operation the system will be in partial vacuum and the reservoir will serve as a buffer tank where gas can accumulate, and limit the pressure rise, before being pumped out of the system. To extend the testing time, a pump should be used to remove the gas from the reservoir.

Conclusions & Recommendations

The preliminary design and analysis of a supersonic nozzle was carried out. A method of characteristics code was developed to generate nozzle contours in two dimensions. An error analysis was performed on the method of characteristics code to show a converged shape is obtained. The generated nozzle contours were then analyzed for design performance. A mesh resolution study was conducted to show mesh independence. The calculated Mach number and pressure distributions were compared with the isentropic solutions to confirm a correct solution is obtained. A brief overview of the other wind tunnel elements was also provided.

Much work remains to be done before a Mach 10 wind tunnel can be constructed and operated and the following tasks will need to be accomplished. A more robust method of characteristics code will be required to design high Mach number nozzles where a gradual expansion is required, and boundary layer growth is accounted for. The inclusion of boundary layer effects requires iteration between the design code and viscous flow simulation. The pressure and temperature along the nozzle wall during operation must be understood to allow for proper construction. This can be accomplished through use of solid mechanics computational tools that can account for the flow pressure loads in a coupled or decoupled approach. Lastly, the nozzle flow field must be analyzed with more accuracy by solving viscous equations, accounting for turbulence effects near the corners, and including the wall heat transfer. Additionally, the flow field of the nozzle during wind tunnel startup could provide useful information for the design of the downstream components.

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